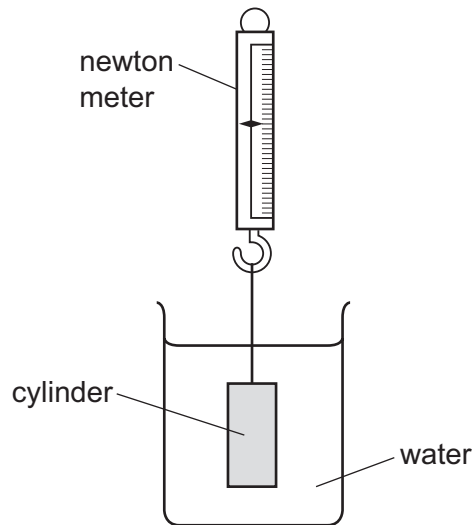


- 14 A uniform cylinder of weight 25.0 N is suspended from a newton meter.

The cylinder is fully submerged in water, as shown.

The reading on the newton meter is 10.0 N.



The water is replaced by a liquid with a density 10% greater than the density of water. The cylinder remains fully submerged.

What is the new reading on the newton meter?

- A 8.5 N B 9.0 N C 11.0 N D 11.5 N
- 15 A student balances a 30 cm ruler on a fulcrum set at the 15 cm mark. She then places a 50 g mass on the 23 cm mark and a 20 g mass on the 11 cm mark, as shown.



Which mass should she place on the 7 cm mark to restore the balance?

- A 30 g B 40 g C 47 g D 133 g
- 16 A submarine is at a depth of 130 m below the surface of the sea.

The pressure on the submarine due to the sea water is p .

The submarine sinks to a depth of 260 m below the surface.

Assume that the density of sea water is constant.

What is the **difference** between the pressures on the submarine due to the sea water at the two depths?

- A 0 B $0.5p$ C p D $2p$